Delta Hedging a Short SPY Straddle: Implied Against Realised Volatility

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1 The Core Idea

A short option that is delta hedged every day does not depend on where the market goes. It depends on one comparison: was the volatility it was sold at (implied) higher or lower than the volatility that actually happened (realised)?

The notebook tests this twice. First on simulated prices, where the true volatility is known. Then on real SPY data from 2010 to 2025, with VIX (Cboe methodology) as the implied volatility.

- **Why it matters:** an option seller earns the volatility risk premium only if implied volatility exceeds realised volatility on average, and loses heavily when it does not.
- **Why the input matters:** the real-data profit exists when the straddle is sold at the VIX level and disappears at a lower at-the-money level.

2 Why It Matters

- **Direction drops out, volatility remains.** After hedging, the trade P&L has a correlation of 0.78 with implied minus realised variance ($R^2 = 0.61$). A desk that sells options without a view on direction is taking a position on volatility.
- **The same trade gives opposite answers.** Sold at VIX, the average trade earns +0.79% of spot. Sold at 0.8×VIX it earns −0.03%. The whole result sits in one input that no one observes in this data set.
- **Wins are small, losses are large.** 83% of trades make money, by 1.21% of spot on average. The worst one, opened on 10 March 2020, lost 7.47%, as much as 9.4 average trades earn. A high win rate says little about risk.
- **Hedging is not free.** In the simulation with a cost of 0.1% per trade, going from 21 to 252 rebalances cuts the hedging error by a factor of 3.3 (0.858 to 0.261) but raises the cost from 0.31 to 1.02, roughly with the square root of the trade count.

3 Assumptions

Each line states what the code assumes and which way the result moves if it is wrong.

- **Black-Scholes with one volatility.** The straddle is priced (Black and Scholes, 1973) and hedged at the same σ for its whole life, with no skew, no smile and no re-marking. If the true implied volatility moves after the sale, the hedge delta is stale and the error grows.

- **VIX stands for the implied volatility.** VIX on the day the trade opens (its close, as for the SPY price) times a scale of 1.0 or 0.8. VIX is a variance swap rate, above the ATM volatility when skew is negative, so a scale of 1.0 overstates the premium. The scale of 0.8 is a judgement: break-even is at 0.80, and 0.9 would give +0.38%.
- **Constant Black-Scholes inputs.** The rate is the 13-week T-bill yield divided by 100, used as a continuous rate and held constant over the trade. The dividend yield is the trailing 252-day dividends over spot at the start. A different rate or yield shifts the premium slightly, not the conclusion.
- **Daily hedging at the close, 21 steps.** The option lasts 21 trading days ($T = 21/252$) and the hedge is reset at each close. Gaps between closes are not hedged. Hedging less often than daily would add noise but not bias.
- **Costs.** The simulation charges 0.1% of notional per share trade, the real-data run 1 bp. The opening hedge is charged, the liquidation at expiry is not, and the option bid-ask spread is ignored. A half-spread of 0.5 vol point on the option would remove about 0.11% of spot per trade (0.23% per vol point).
- **Non-overlapping monthly trades.** The first trade opens on 4 January 2010 and a new one opens every 21 trading days, 191 in all. A different start day moves the sample mean by about 0.1 point of spot (Section 7).
- **Geometric Brownian motion in the simulation.** Constant volatility of 20%, no jumps, no clustering, risk-neutral drift of 3%, 5,000 paths and one random seed (42). A tail loss like March 2020 cannot occur in it.
- **European treatment of American options.** Early exercise matters little for a one-month ATM straddle with a dividend yield of about 1.7%.
- **P&L in percent of spot.** Margin, capital and funding of the short option are left out, so the numbers are not returns on capital.

4 Method

4.1 Price and delta

European Black–Scholes with a dividend yield q. With $d_1 = \dfrac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$:

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2), \qquad P = Ke^{-rT}N(-d_2) - Se^{-qT}N(-d_1).$$

The straddle is $C + P$ at the same strike, and its delta is $e^{-qT}\,(N(d_1) + N(d_1) - 1)$. At the money the premium is close to $0.8\,S\sigma\sqrt{T}$, so it grows with volatility, and the straddle delta is small and positive because of the drift term.

4.2 The hedging engine

The straddle is sold at the implied volatility σ and hedged at the same σ. At each date the position holds Δ_i shares, the delta of the straddle. Between dates, cash earns the rate r and the

shares pay dividends. Every share trade costs a fraction c of its value. At expiry the straddle payoff is paid and the shares held (Δ_{n-1}) are sold:

$$\text{P\&L} = \text{cash}_T + \Delta_{n-1}\, S_T - |S_T - K|.$$

The result is the P&L of the hedged position, reported as a percentage of the strike (spot at inception).

The intuition is that the hedge buys shares after a rise and sells them after a fall, so a path that oscillates costs money (gamma), while the passage of time pays the seller (theta). The seller breaks even on a day when the move is about $\sigma/\sqrt{252}$. If the straddle is sold at σ_{imp} and the market realises σ_{real}, the mean P&L is close to the difference in premium, $V(\sigma_{imp}) - V(\sigma_{real})$, whatever the number of rebalances. The number of rebalances sets the spread around that mean.

4.3 Simulation and real data

- **Simulation:** geometric Brownian motion, $S_0 = 100$, one-month ATM straddle ($T = 21/252$), $r = 3\%$, true volatility 20%, 5,000 paths, seed 42. The straddle is sold at 20% or 25%, and hedged 1, 5, 21, 63 or 252 times. Costs are 0 or 0.1% per trade.
- **Real data:** every 21 trading days, sell a 21-day ATM straddle on SPY and hedge it daily at the close, 191 non-overlapping trades. σ is VIX at the start date times a scale (1.0 or 0.8). The rate is the 13-week T-bill (`^IRX`), the dividend yield is the trailing 252-day dividends over spot, and the cost is 1 bp per share trade.
- **Realised volatility:** standard deviation of the daily log returns over the trade, times $\sqrt{252}$.

5 Method Comparison

Set-up	Strength	Weakness
Simulation, true volatility	Known answer; checks the engine	No jumps, no clustering
Real data, sold at VIX	Real paths; large t-statistic (8.4)	VIX is not an option price
Real data, sold at 0.8×VIX	Closer to an ATM level; edge gone	The factor is a judgement

The simulation is the right place to check that the hedging code is correct: the mean is zero at the true volatility and equals the premium gap at a wrong one. The two real-data runs bracket the answer: the first is an upper bound on what a seller could have sold at, the second is a more cautious level. Only real option quotes would settle it.

6 Results

Simulation. Realised volatility is 20%, and the price of the straddle at 25% minus its price at 20% is 1.149. The mean P&L sold at 20% without cost stays between -0.034 and 0.026 for every frequency. The last column is the Derman–Kamal leading-order formula for the straddle, $2\sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{N}$ with call vega $\mathcal{V} = 11.49$, added for comparison (it is not in the code).

	Sold at 20%			Sold at 25%, mean P&L	
Rebalances	Std	Mean, cost 0.1%	Formula sd	No cost	Cost 0.1%
1	3.510	0.020	4.07	1.177	1.172
5	1.702	−0.160	1.82	1.113	1.001
21	0.858	−0.308	0.89	1.129	0.871
63	0.513	−0.513	0.51	1.159	0.704
252	0.261	−1.017	0.26	1.166	0.256

- **True volatility:** when the straddle is sold at the true volatility, the hedged P&L averages zero. Its standard deviation falls from 3.51 to 0.26 as the number of hedges goes from 1 to 252, close to the formula once N is 21 or more.
- **Overpriced volatility:** sold at 25% against 20% realised, the seller keeps about 1.1 to 1.2, close to the price gap of 1.149, at every frequency. The differences between 1.113 and 1.177 are within sampling error (the standard error of a mean is the standard deviation over $\sqrt{5000}$: 0.05 for one rebalance, under 0.01 for 252).
- **Costs:** with 0.1% per trade, 252 rebalances cut the edge from 1.166 to 0.256.

The cost column at 20% shows that cost grows more slowly than the number of trades: from 21 to 252 rebalances the number of trades is 12 times larger and the cost is 3.3 times larger (1.017/0.308), a little below the $\sqrt{12} = 3.46$ of an exact square-root law. The hedge trades smaller amounts when it trades more often, so the total volume grows with $\sqrt{N}$. The error falls with $\sqrt{N}$ while the cost rises with $\sqrt{N}$, so there is a frequency that balances the two. The simulation does not show the behaviour in a jump, since its paths have none.

SPY, 2010 to 2025 (4,024 trading days, 191 trades):

	Sold at VIX	Sold at 0.8×VIX
Mean P&L per trade (% of spot)	+0.79	−0.03
Winning trades	83%	62%
Worst trade	−7.5% (10 Mar 2020)	−8.2% (10 Mar 2020)

VIX averaged 18.4% against 14.5% realised. The correlation between trade P&L and (implied variance − realised variance) is 0.78 ($R^2 = 0.61$).

Interpretation. The VIX is 1.27 times the realised volatility on average, and realised volatility is above VIX in only 17% of the months. That is the volatility risk premium, and it explains the +0.79%: a gap of 3.9 vol points times about 0.23% of spot per vol point is 0.90%, the difference between the premiums at 18.4% and 14.5%. With 0.8×VIX the implied volatility averages about 14.7%, which is almost the realised value, and the premium vanishes: the mean falls to −0.03% with a t-statistic of −0.29. Both runs have a negative skew (−2.15 and −2.72) and a median above the mean (0.95% and 0.17%): most months pay a little, a few months give back several months of profit. The worst trades are 10 March 2020 (realised 87% against VIX at 47%), 5 January 2018 (−5.4%, realised 20% against implied 9%), 3 August 2011 and 14 March 2025. The table does not tell whether an option seller could have earned the +0.79% in practice. It is a measure of the VIX level against realised volatility, not of a tradable edge.

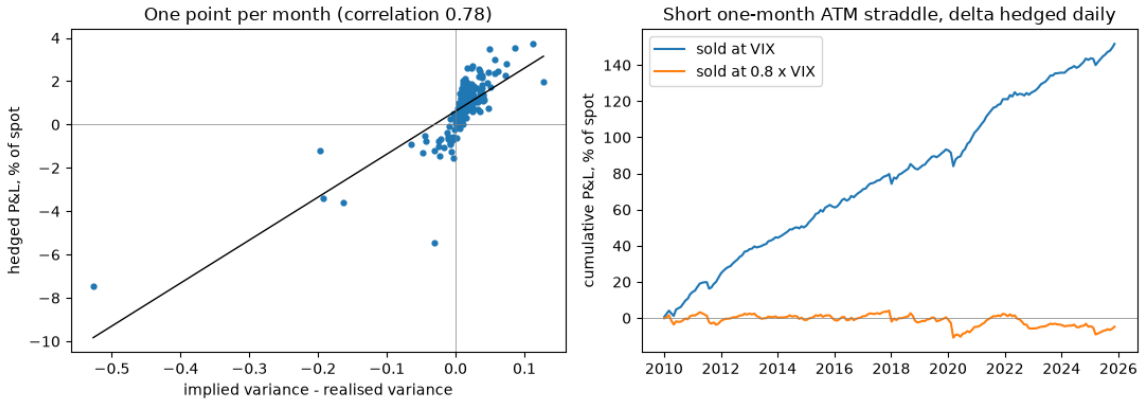


Figure 1: Left: hedged P&L against implied minus realised variance, one point per trade. Right: cumulative P&L of the two runs.

7 Worked Example

Premium and delta (illustrative): $S = K = 100$, $T = 21/252$, $r = 3\%$, $q = 0$, $\sigma = 20\%$.

$$C = 2.4271, \quad P = 2.1774, \quad C + P = 4.6045,$$
$$\Delta_{straddle} = 0.5288 + (-0.4712) = 0.0575,$$

so the hedge holds 0.0575 shares per straddle sold. A quick check on the premium: an ATM straddle is worth about $0.8\, S\sigma\sqrt{T} = 0.8 \times 100 \times 0.20 \times \sqrt{21/252} = 4.62$.

One hedging step (illustrative): the price rises 1% to 101 after one day. The straddle delta becomes 0.1951, so the seller buys $0.1951 - 0.0575 = 0.1376$ shares at 101, a cost of 0.0014 at 1 bp. If the price falls 1% instead, the delta is -0.0860 and the seller sells 0.1435 shares at 99. The hedge buys high and sells low, and the cost of that is what theta has to cover.

Edge from overpriced volatility. At $\sigma = 25\%$ the premium is 5.7532, so the gap is $5.7532 - 4.6045 = 1.1487$. The simulation mean (1.113 to 1.177 without costs) is close to this number.

Is the edge real? The t-statistic on the 191 trades is recomputed from the notebook's trade table. Sold at VIX: mean 0.794%, standard deviation 1.312%.

$$\text{s.e.} = 1.312/\sqrt{191} = 1.312/13.82 = 0.0949,$$
$$t = 0.794/0.0949 = 8.37.$$

Sold at 0.8×VIX: mean -0.025%, standard deviation 1.208%, so $t = -0.025/(1.208/13.82) = -0.29$. The t-statistic of 8.4 is for a mean of a skewed variable with 191 months of one path of history, so it is a measure of the gap between VIX and realised volatility, not a guarantee.

Sensitivity. Rerunning the notebook code:

- *Scale of VIX.* Mean P&L per trade for scales of 0.70, 0.75, 0.80, 0.85, 0.90, 0.95 and 1.00: -0.43, -0.23, -0.03, $+0.18$, $+0.38$, $+0.59$ and $+0.79$ (% of spot), with win rates of 42%, 51%, 62%, 68%, 74%, 80% and 83%. Each 5% of scale is worth about 0.2% of spot per trade, and the break-even is 0.80.
- *Start day.* Moving the first trade by 4, 8, 12, 16 and 20 trading days gives a mean of $+0.88$, $+0.84$, $+0.84$, $+0.81$ and $+0.77$ at VIX, and $+0.03$, -0.01, -0.02, -0.03 and -0.06 at 0.8×VIX. The sign at 0.8×VIX is not stable and the VIX result is.

- *Without the worst trade.* Dropping the 10 March 2020 trade raises the mean to +0.84% at VIX and to +0.02% at 0.8×VIX.
- *By year.* At VIX the mean is positive in all 16 calendar years (from +0.21% in 2018 to +1.55% in 2021). At 0.8×VIX it is negative in 8 of 16 (2011, 2013, 2014, 2018, 2020, 2022, 2024, 2025), with a low of −0.68% in 2022.
- *Volatility doubles* (illustrative, 40% instead of 20%): the premium rises from 4.60 to 9.20, and the break-even daily move rises from 1.26% to 2.52%.

8 Limits and Extensions

- **No option prices:** implied volatility is approximated from VIX. The factor 0.8 is a judgement, not a measurement, and the conclusion depends on it.
- **VIX is not ATM volatility:** VIX is a 30-day variance swap rate. With a negative skew it sits above the at-the-money volatility.
- **Simplified market:** the bid-ask spread of the options is ignored, SPY options are priced as European, and the hedge uses closing prices only.
- **Tail dominated:** the losses are concentrated in a few months when realised volatility jumped (March 2020). A sample of 191 trades holds few such events. These are model outputs, not recommendations.

Three extensions would address these limits. Real SPY option quotes would replace VIX by the actual ATM price and give a bid-ask cost. A GBM with jumps or stochastic volatility in the simulation would show how the hedging error behaves in a tail. Reporting the P&L per unit of capital at risk, using a margin model, would make the trades comparable with other strategies.

9 Conclusion

Delta hedging turns an option position into a bet on the gap between implied and realised volatility, and on SPY that gap is positive at the VIX level and nil at a plausible ATM level. A risk manager should therefore ask which implied volatility a trade was actually sold at, and size the position for the March 2020 month, not for the average one.

References

Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654. `https://doi.org/10.1086/260062`

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